Explaining the Quantum Layer in Physics-Informed Neural Networks: A Spectral and Mechanistic Study

(Extended / Working Report with Roadmap)

WISER Global Quantum+AI Program 2026 — BQP Challenge
Quantum-Assisted Physics-Informed Neural Networks for CFD

Draft — July 2026

Abstract

This is the extended working version of the QAPINN-BQP report. It contains the full technical content of the submission draft — a verified parameter-matched test bench, a five-dimension explainability suite, and a Tier-1 ablation on Burgers — and additionally documents the *complete experimental roadmap*, the statistical work required to promote our current single-seed hypotheses to claims, and an explicit account of everything currently *left out* of scope. Headline hypotheses (single seed, one undertrained PDE): a QAPINN's advantage tracks the frequency-setting knobs (depth, qubits, encoding scale), not *entanglement* at fixed band; and against a structurally-identical classical-front-end control the advantage is *bandwidth-gated* while the apparent classical-side effects (neuron diversity, gradient amplification) do *not* survive — what remains is a bandwidth-gated advantage and a barren-plateau signature in qubit count. The pipeline is independently verified (8/8 checks); the conclusions are not yet, and the path to establishing them is the subject of Sections 6–7.

Contents

1 Introduction **2**

2 Background and Related Work **2**

3 Methodology **2**
3.1 Test problems 2
3.2 Models and XAI suite 3
3.3 Verification 3

4 Experiments and Results **3**
4.1 Baseline and spectra 3
4.2 Tier-1 ablation 3
4.3 Tier-3 ablation: qubits, measurement, encoding scale 3
4.4 How the quantum layer changes the classical network 4
4.5 Discussion 5

5 Current Limitations **6**

6 Future Work and Experimental Roadmap **6**
6.1 Tier 2a — Convergence / iteration analysis (highest priority) 6
6.2 Tier 2b — Multi-seed robustness (error bars) 6
6.3 Tier 2c — The classical-front-end control 7
6.4 Tier 3 — Remaining ablation axes and the bandwidth continuum 7
6.5 Tier 4 — Real CFD and mathematical depth 7
6.6 The deliverable — design methodology 8

7 What Is Currently Left Out (Scope Boundaries) **8**

8 Conclusion **9**

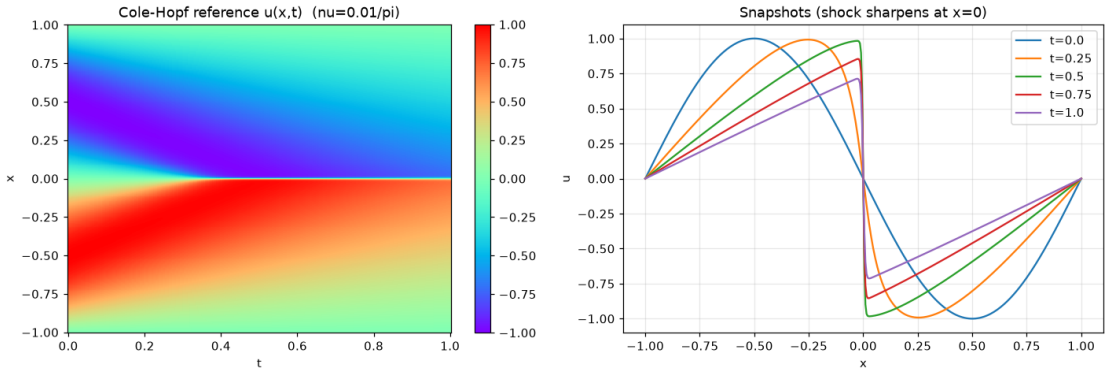


Figure 1: Exact Cole–Hopf Burgers reference: field (left) and snapshots (right), showing the shock at $x = 0$.

1 Introduction

Physics-Informed Neural Networks (PINNs) [1] make the differential equation the loss. A Quantum-Assisted PINN (QAPINN) [4] replaces the first hidden layer with a variational quantum circuit (VQC), and is reported to cut trainable parameters at maintained accuracy. The BQP challenge asks us to *explain* the learning dynamics — when, why, and how the quantum layer changes learning — and to derive a problem-specific circuit-design methodology.

We adopt the dominant theoretical lens for VQCs (a data-encoding circuit is a truncated Fourier series [2]) and test whether it suffices to explain a QAPINN's behaviour, and how the quantum layer changes the classical network it is attached to. Contributions and results mirror the submission report; the extended value here is the roadmap (Sec. 6) and scope boundaries (Sec. 7).

2 Background and Related Work

A VQC with data-encoding gates is a partial Fourier series $f(x) = \sum_{\omega\in\Omega} c_\omega e^{i\omega x}$: the frequency set Ω is fixed by the encoding (with L re-uploads, $\Omega = \{-L, \dots, L\}$), the coefficients c_ω by the trainable gates [2]. Mhiri et al. [5] show the coefficients are additionally constrained and can vanish with qubit count — frequencies are necessary but not sufficient. Differentiable quantum circuits solve DEs with analytic derivatives [3]; hybrid QAPINN/QCPINN designs [4, 6] report PDE-dependent parameter reduction (quantum sometimes *loses*, e.g. cavity flow); a "black-hole" collapse to the trivial solution is a distinct trainability pathology [7]. No validated mechanism explains *when/why* the quantum layer helps — the gap targeted here.

3 Methodology

3.1 Test problems

Same domain, IC $u(x, 0) = -\sin(\pi x)$, and BCs for both PDEs, so only the operator differs: Burgers $u_t + uu_x = \nu u_{xx}$ ($\nu = 0.01/\pi$, shock $\to$ broadband) with an exact Cole–Hopf reference (Fig. 1); heat $u_t = \alpha u_{xx}$ ($\alpha = 1/\pi^2$, single mode $u = -\sin(\pi x)e^{-t}$).

Check	Method	Result
Burgers residual	vs analytic	1.2×10^{-7}
Heat residual	exact $\to$ residual	6.0×10^{-8}
Autograd u_x	vs finite diff	match
VQC = Fourier, max freq = depth	$L{=}2 \to 2,\ L{=}4 \to 4$	pass
relative-L_2	exact $\to 0$	4.0×10^{-8}
Matched params	all configs	121
Reproducibility	same seed	bit-exact

Table 1: Correctness audit (8/8).

3.2 Models and XAI suite

Classical PINN (tanh MLP) vs QAPINN (fixed coord-map $\to$ VQC $\to$ classical post-net), compared at matched parameters. Baseline VQC: 4 qubits, depth 3, *RY* re-uploading, linear CNOT chain, $\langle Z_i \rangle$ measurement, post-net [16] (121 params, 24 quantum). Five XAI dimensions (spectral/F-principle; neurons; VQC internals incl. entanglement entropy; loss landscape; generalisation) plus training-time spectrum and per-layer/per-qubit gradient-variance probes.

3.3 Verification

All eight independent correctness checks pass (Table 1); notably, the VQC is confirmed to be a truncated Fourier series with maximum frequency equal to the re-upload depth.

4 Experiments and Results

4.1 Baseline and spectra

At 121 matched params (Adam, 2000 iters, seed 0): heat solved by both (classical 8.3×10^{-3}, QAPINN 5.9×10^{-3}); Burgers undertrained (classical 0.433, QAPINN 0.472). Spectra (Fig. 2): heat is a single $k{=}1$ mode; Burgers is broadband and both models undershoot the high harmonics (depth-3 VQC capped near $k \approx 3$).

4.2 Tier-1 ablation

Depth = bandwidth (Fig. 3a): QAPINN accuracy improves monotonically ($0.57 \to 0.42$), matched classical stays flat; realised max harmonic grows with depth; QAPINN overtakes classical at depth ≥ 3. **Entanglement at fixed band** (Fig. 3b): accuracy barely moves (0.456–0.488), *more* entanglement slightly *worse*, none is best — contradicting entanglement-driven expressibility. **Post-net width**: the VQC needs a readout but beats matched classical for any post-net $\geq$ [8].

4.3 Tier-3 ablation: qubits, measurement, encoding scale

The remaining knobs (Table 3, Fig. 8) reinforce the picture. **Qubits** behave like depth: accuracy improves monotonically ($0.501 \to 0.443$), QAPINN overtakes matched classical at ≥ 4 qubits. **Measurement:** single $\langle Z_0 \rangle$ bottlenecks (0.591); multi = ZZ (0.472) — readout width matters, correlator type does not. **Encoding scale:** optimum near 1.0, catastrophic collapse to the trivial solution at 2π — the black-hole pathology as an ablation.

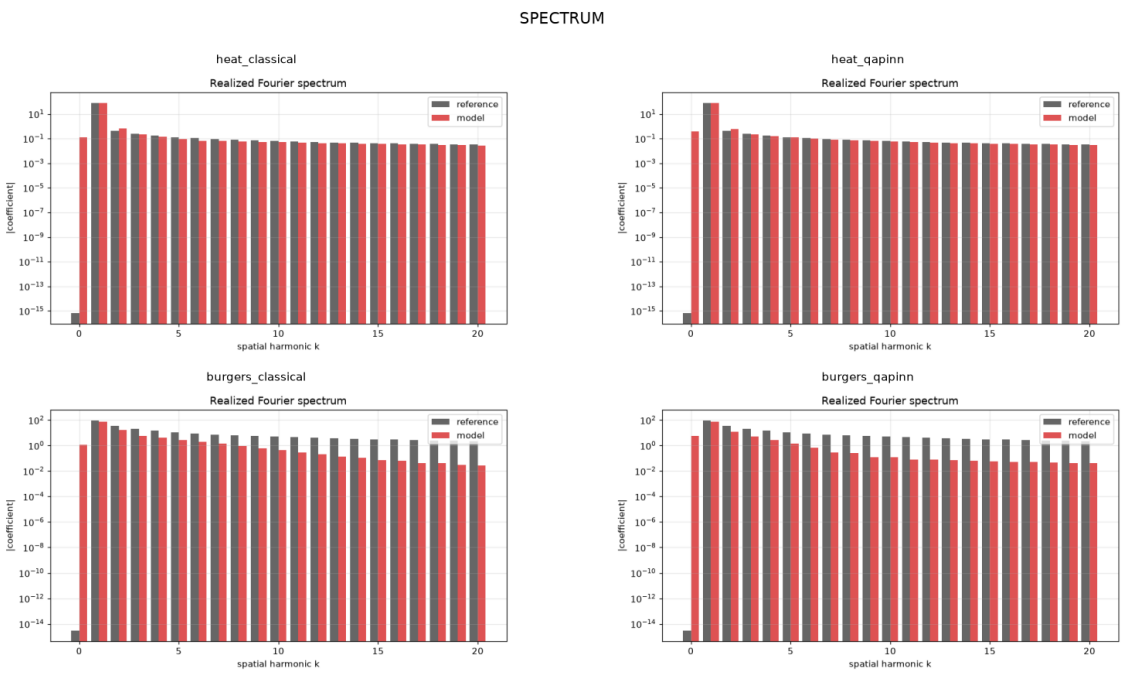
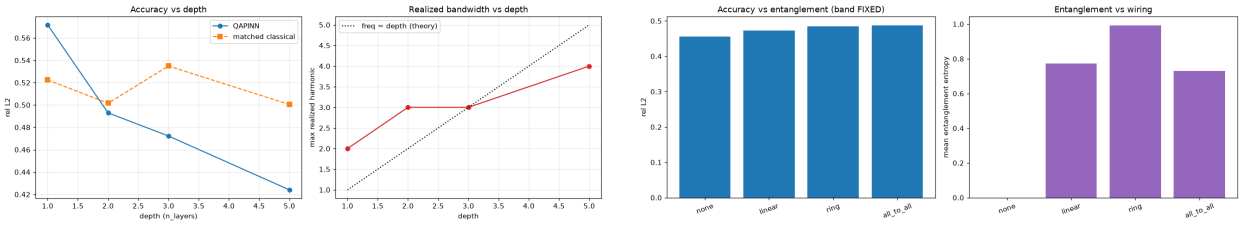


Figure 2: Realised spectra (grey reference, red model). Burgers (bottom): both models undershoot the high-frequency shock tail.



(a) Depth.

(b) Entanglement (band fixed).

Figure 3: Tier-1 ablations on Burgers.

Five knobs split cleanly: *bandwidth* knobs (depth, qubits, encoding scale) carry the advantage up to a trainability ceiling; *coefficient/readout* knobs (entanglement, correlator measurement) do not — the bandwidth-vs-trainability squeeze, across five knobs.

4.4 How the quantum layer changes the classical network

A naïve comparison against a plain PINN suggested the quantum layer makes the downstream neurons more diverse and drives larger gradients, but that comparison is confounded (different post-net shape). We instead use a *structurally identical* classical-front-end control (CFE): the same post-net fed by $\text{Linear}(2 \to w) + \tanh$ instead of the VQC. Against this fair control (Table 4), the effects do *not* hold: accuracy is bandwidth-gated (QAPINN ties at low depth/qubits, wins at depth 5 / 6 qubits / heat); neuron diversity is inconsistent (classical front-end more diverse on the depth axis, quantum more on qubits); and gradient amplification reverses at 5–6 qubits, where the QAPINN gradient variance shrinks with qubit count (a barren-plateau signature). What survives: a bandwidth-gated advantage and a barren-plateau signature — the classical-side "mechanism" is

Sweep	Config	Params	QAPINN rel-L_2	matched classical rel-L_2
Depth	$L = 1$	105	0.572	0.522
	$L = 2$	113	0.493	0.502
	$L = 3$ (baseline)	121	0.472	0.535
	$L = 5$	137	**0.424**	0.500
Entanglement (band fixed)	none	121	**0.456**	0.535
	linear	121	0.472	0.535
	ring	121	0.485	0.535
	all-to-all	121	0.488	0.535
Post-net	[] (linear)	29	0.544	0.533
	[8]	73	0.478	0.533
	[16] (baseline)	121	0.472	0.535
	[32]	217	**0.466**	0.505

Table 2: Complete Tier-1 ablation results (single seed, Burgers, 2000 Adam iters). QAPINN improves with depth and post-net width and overtakes the matched classical control; at fixed band, entanglement does not help (*none* best).

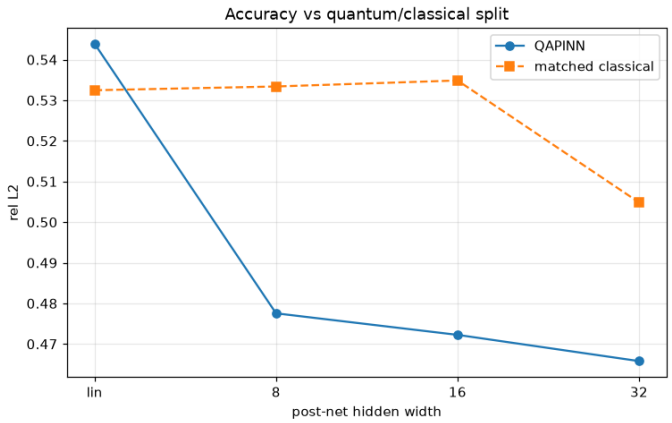


Figure 4: Post-net width sweep: QAPINN vs matched classical. The VQC needs a small readout but beats matched classical for any post-net $\geq$ [8].

not a clean effect.

4.5 Discussion

The evidence is coherent: the quantum layer's behaviour is a *bandwidth* phenomenon (tracks depth/qubits/scale, not entanglement at fixed band, needs but is not subsumed by a classical readout, holds up to a trainability ceiling). Against a fair control the advantage is *bandwidth-gated* and the apparent classical-side mechanism does *not* survive — what remains is a bandwidth-gated advantage and a barren-plateau signature in qubit count.

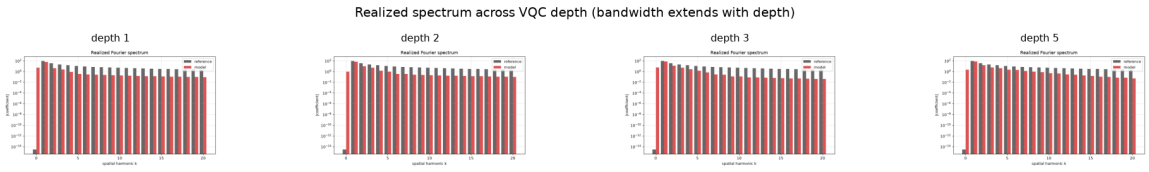


Figure 5: Realised spectrum (grey reference, red model) across VQC depth — the model captures more of the high-frequency shock tail as depth grows.

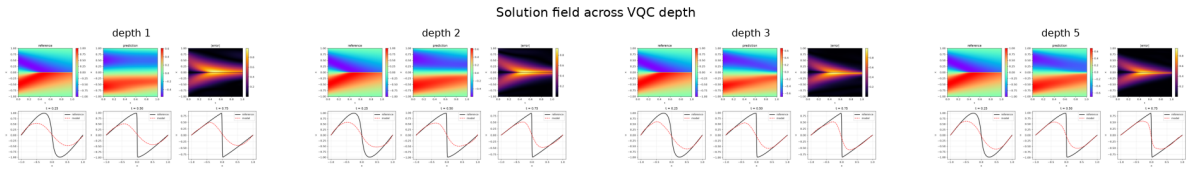


Figure 6: QAPINN solution field across VQC depth.

5 Current Limitations

Hypotheses, not claims: (1) single seed — pairwise gaps (∼7%) are within plausible init noise; (2) undertrained, one PDE (Burgers 0.42–0.57); (3) a structural confound in the diversity comparison (16 vs 30 neurons), mitigated by the fixed-post-net depth sweep; (4) simulation only (no shot noise / hardware). Code verified; conclusions not yet.

6 Future Work and Experimental Roadmap

The single most important next step follows from the "undertrained" limitation: *establish convergence before spending compute on statistics.* The revised priority order is therefore convergence, then seeds, then breadth.

6.1 Tier 2a — Convergence / iteration analysis (highest priority)

Because all current results are undertrained, every ordering may be a transient. Iterations is nearly free as an ablation: checkpoints and the loss history from a single long run yield every iteration point. Plan: long-train the headline configs (baseline QAPINN, entanglement-none QAPINN, matched classical) to ∼8000 iters with dense checkpoints, and re-evaluate all metrics (accuracy, realised spectrum, entanglement, classical-side diversity/gradients) along the trajectory. Questions: do the depth and entanglement orderings persist at convergence? Does the slower-training QAPINN (its k=1 error halves at iter 550 vs 75 for classical) catch up, plateau higher, or overtake? Do the classical-side effects grow or fade? Cost: ∼5 h.

6.2 Tier 2b — Multi-seed robustness (error bars)

Re-run the surviving headline configs across 3–5 seeds to obtain mean ± std, and accept a difference only if it exceeds the seed spread. Priority: the entanglement-does-not-help result (publishable if it survives) and the classical-side diversity/gradient effects. Cost: ∼10 h.

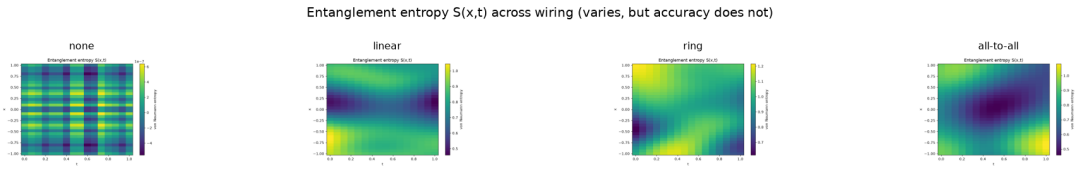


Figure 7: Entanglement entropy $S(x,t)$ across entangling patterns: entropy varies strongly while accuracy does not (Table 2).

Sweep	Config	Params	QAPINN rel-L_2	matched cls. rel-L_2
Qubits	3	99	0.501	0.470
	4 (base)	121	0.472	0.535
	5	143	0.468	0.541
	6	165	**0.443**	0.465
Measure	single	73	0.591	0.533
	multi (base)	121	0.472	0.535
	ZZ	169	0.472	0.515
input_scale	0.5	121	0.513	0.535
	1.0 (base)	121	**0.472**	0.535
	π	121	0.516	0.535
	2π	121	**1.000**	0.535

Table 3: Tier-3 ablation results (single seed).

6.3 Tier 2c — The classical-front-end control

To remove the last confound from the "diversity" finding, add a control with an *identical* post-net but a classical $2 \to 4$ front-end, and compare its post-net directly to the QAPINN's. This isolates "quantum vs classical front-end effect on the same classical net." Cost: negligible (classical).

6.4 Tier 3 — Remaining ablation axes and the bandwidth continuum

- **ν-sweep (bandwidth continuum):** tune Burgers viscosity from smooth to sharp and plot advantage vs solution bandwidth — the continuum that turns two discrete regularity points (heat, Burgers) into a scaling relation. This directly tests whether bandwidth is a *sufficient statistic* for advantage.
- **Qubits** (3–6): more bandwidth + Hilbert space (changes params; needs re-matching + barren-plateau watch).
- **Measurement:** single / multi / ZZ, and the not-yet-implemented *probability vector* readout that the challenge explicitly requests.
- **input_scale** and **loss weighting:** secondary bandwidth / optimisation levers (the latter drove the earlier full-quantum collapse).
- **Encoding scheme:** the coordinate→qubit map (round-robin vs split vs all-on-all).

6.5 Tier 4 — Real CFD and mathematical depth

- **Lid-driven cavity (2-D incompressible Navier–Stokes):** the real-world CFD case the judges reward; the PDE-agnostic bench already supports multi-output, multi-equation systems. Literature suggests quantum may *lose* here — a valid "what is lost" finding.

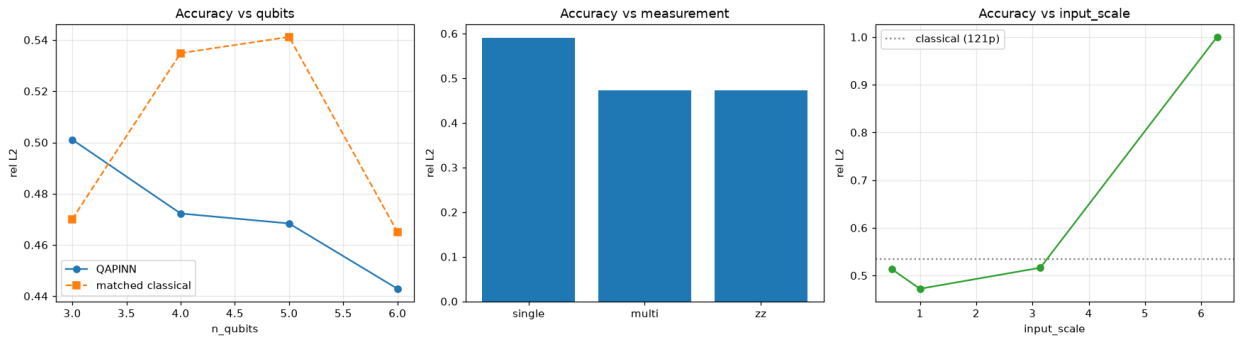


Figure 8: Tier-3 sweeps: qubits (bandwidth-like), measurement (readout width vs type), encoding scale (optimum + 2π collapse).

Comparison	rel-L_2 (Q / C)	diversity (Q / C)	grad-var $\times 10^{-3}$ (Q / C)
Burgers depth-3	0.472 / 0.474	0.269 / **0.286**	**4.1** / 0.62
Burgers depth-5	**0.424** / 0.474	0.236 / **0.286**	**7.1** / 0.62
Heat	**0.006** / 0.013	**0.272** / 0.239	**3.2** / 0.47
Burgers 3 qubits	0.501 / 0.496	**0.266** / 0.241	**1.2** / 0.26
Burgers 5 qubits	0.468 / 0.464	**0.312** / 0.265	1.1 / **5.2**
Burgers 6 qubits	**0.443** / 0.467	**0.341** / 0.269	0.72 / **1.9**

Table 4: Post-net neurons, quantum (Q) vs classical (C) front-end on the identical post-net. Bold = larger of each pair. Single seed.

- **Fourier coefficient reachability (Mhiri):** measure realised *coefficients* vs the reference, not just the frequency support, to test whether the depth-5 bandwidth shortfall (max freq 4, not 5) is the coefficient constraint biting.
- **Barren-plateau / trainability study:** systematic gradient-variance vs qubits/depth over many random initialisations; relation to the "black-hole" collapse.

6.6 The deliverable — design methodology

The end goal (challenge Task 4) is a validated chain: *cheap proxy* (Reynolds/ν, or a coarse classical/FEA solve) $\rightarrow$ estimated solution bandwidth $\rightarrow$ prescribed circuit (qubits, depth, encoding). Validation uses exact solutions; deployment uses the proxy, resolving the circularity of "needing the solution to size the circuit."

7 What Is Currently Left Out (Scope Boundaries)

For transparency, the present study explicitly does *not* yet include:

- **Statistics:** no repeated seeds, no error bars, no significance tests — every result is a single point estimate.
- **Convergence:** nothing trained to convergence; all Burgers results are Adam-only and undertrained.
- **Breadth of PDEs:** only Burgers and heat; no real CFD (cavity), no ν-continuum.
- **Full VQC knob set:** qubits, measurement (incl. probability vector), input_scale, encoding scheme, and loss weighting are not yet ablated; Tier 1 covered only depth, entanglement, and

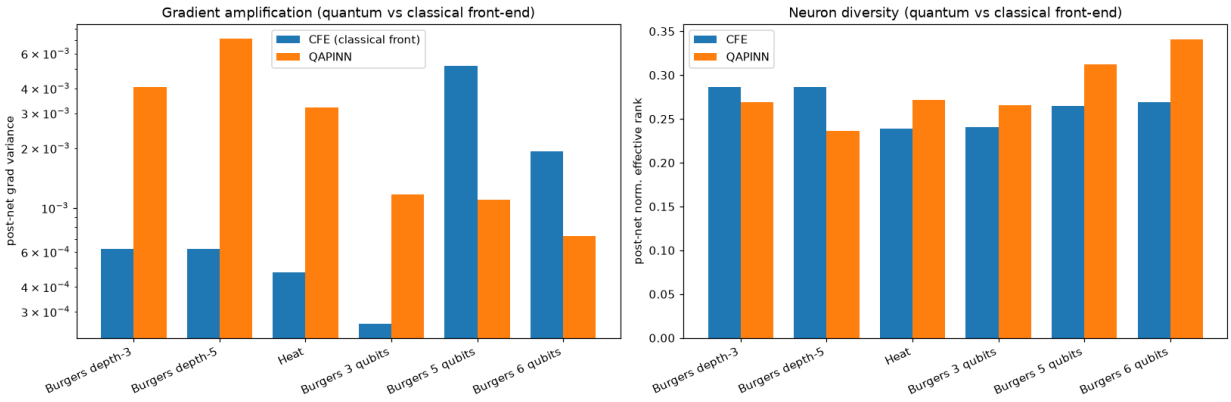


Figure 9: Quantum vs classical front-end on the identical post-net: gradient variance (left) and neuron diversity (right). Neither effect is robust across knobs.

Tier	Content	Est. compute	Promotes
2a	convergence (iterations)	∼5 h	undertraining → converged
2b	3–5 seeds on headlines	∼10 h	hypotheses → error-barred
2c	classical-front-end control	∼0	removes diversity confound
3	ν-sweep + remaining knobs	∼15 h	mechanism → sufficiency
4	real CFD + coefficient/BP analysis	TBD	breadth + math rigour

Table 5: Experimental roadmap with rough compute cost (single laptop, `default.qubit`). `lightning.qubit` may accelerate the simulation but currently breaks the higher-order autograd needed for u_{xx}.

post-net width.

- **Coefficient-level Fourier analysis:** we measure realised frequency *support*, not the realised *coefficients* vs their reachable set (Mhiri).
- **Trainability theory:** the black-hole collapse and barren plateaus are discussed but not systematically measured on QAPINNs.
- **Hardware / noise:** statevector simulation only; no shot noise, no device runs; "parameter reduction" is not "resource reduction" (∼85× slower in simulation).
- **Design methodology:** the proxy→bandwidth→circuit rule is designed but not yet validated end-to-end.

8 Conclusion

We have a verified bench, a five-dimension XAI suite, and a coherent set of single-seed hypotheses: the quantum layer's behaviour is a bandwidth phenomenon (not entanglement). Against a structurally-identical classical-front-end control the advantage is bandwidth-gated, and the apparent classical-side mechanism does not survive — what remains is a bandwidth-gated accuracy advantage and a barren-plateau signature in qubit count. The immediate next step is the convergence (Tier 2a) analysis, followed by multi-seed error bars (Tier 2b), after which the surviving hypotheses become claims and the design methodology can be validated.

References

[1] M. Raissi, P. Perdikaris, G. E. Karniadakis. *Physics-informed deep learning (Part I)*, 2019.

[2] M. Schuld, R. Sweke, J. J. Meyer. *Effect of data encoding on the expressive power of variational QML models*. Phys. Rev. A **103**, 032430 (2021).

[3] O. Kyriienko, A. E. Paine, V. E. Elfving. *Solving nonlinear DEs with differentiable quantum circuits*. Phys. Rev. A **103**, 052416 (2021).

[4] J. Shah, R. Lineswala, A. Chopra. *Benchmarking QA-PINN for CFD*. IEEE QCE 2024.

[5] S. Mhiri et al. *Constrained and Vanishing Expressivity of Quantum Fourier Models*. Quantum **9**, 1847 (2025).

[6] A. Farea, M. Khan. *QCPINN: Quantum-Classical PINNs for PDEs*. MLST (2025).

[7] Chen et al. *Quantum PINNs for Maxwell's Equations*. Quantum Machine Intelligence (2026).